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PDA
Parenteral Drug Association

Points to Consider No. 9

Lessons Learned from the COVID-19 Pandemic

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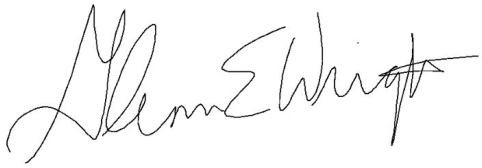
Foreword

Points to Consider (PtC) No. 9 was started during the height of the pandemic. It captures ideas and learnings that were considered important to the pharmaceutical manufacturing industry during a tumultuous period when industry and regulators were looking for, and struggling with, how to move forward. Unlike many of our purely technical documents, this PtC captures a wide variety of topic areas and touches on points normally outside of PDA's normal purview, such as human resources, public policy, and employee transportation. The intent was to provide a document that could be used to prepare for and respond to a similar pandemic if one was to occur in the future. While this document is not intended to cover all possible topics, it assists in ensuring that the industry does not forget the major learnings.

As we close out this PtC activity, PDA would like to sincerely thank all those members from the industry and all of the regulatory authorities who worked so diligently throughout the pandemic to ensure that both new and existing products continued to be produced and made available to patients. The long hours, lost sleep and, in many cases, risk to personal health (especially before the arrival of the vaccine) that was required to keep the industry moving forward will be forever appreciated.



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Introduction

As a result of the COVID-19 pandemic, many industries–the pharmaceutical industry included–went through a substantial disruption. Some of these disruptions included facility shutdowns, unprecedented supply chain challenges, worker availability, and quality system limitations (e.g., physical signatures, investigations, document originals, risk assessments, audits). For many pharmaceutical companies, challenges increased as they raced to provide a vaccine or a therapeutic treatment for COVID-19. Their manufacturing operations were changed over to accommodate the creation of the vaccine and, in order to get the vaccine to patients quickly, many logistical challenges arose.

As the pandemic has progressed, our understanding of the virus, its control, and treatment options have changed and advanced. When reading this document, it is important to remember that its intent is to describe the lessons learned for future benefit, especially at times when the understanding of the virus and treatment options were extremely limited. Even today, as we enter what is believed to be the last phase of the pandemic, moving into an endemic state, it is difficult to comprehend the amount of anxiety and risk that existed during the initial 12 to 16 months. During that time, some types of government-imposed state of emergency were mandated in many countries that restricted the movement of individuals and closed nonessential businesses. Hospitals operated at capacity, a significant number of Covid-related deaths occurred, and a limited amount of personal protective equipment (e.g., mask, gloves) was available.

The specifics of the pandemic or the COVID-19 virus are not detailed here, as that information is widely available. One point that does need to be made is that this pandemic, like the pandemic of 1918, went through waves, and the exact timing of each wave and the number of waves experienced varied by country. Provided below is an example of the COVID-19 waves for the United States **(Figure 1)**, published in the Morbidity and Mortality Weekly Report, that shows the seven-day moving average number of cases in the United States over a two-year period ***(1)***. These waves were not unexpected but how the operation of manufacturing facilities should react to these first waves was not immediately clear; that developed over time. This document intends to convey the understanding and evolution of thought in response to the new data and observations presented as we worked our way through 2019, 2020, 2021, and 2022.

Scope

This document applies lessons learned from the COVID-19 pandemic to pharmaceutical manufacturing. Areas covered include manufacturing, people, quality, regulatory, and supply chain.

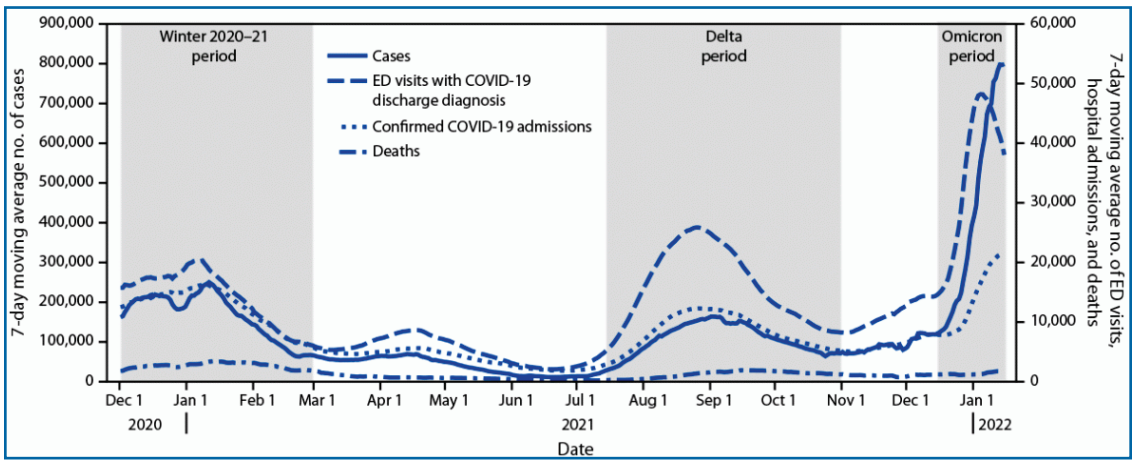

Figure 1 Seven-day moving average number of COVID-19 cases, emergency department visits, hospital admissions, and deaths, December 1, 2020–January 15, 2022 ***(1)***

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I. Manufacturing

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Q 1-1: Are GMP controls sufficient to protect the product?

Background

Due to the novelty of the coronavirus, at the start of the pandemic, the potential impact to the safety and quality of the drugs produced as a result of infected manufacturing employees was uncertain. The virus was found to spread very efficiently, which resulted in a much higher percentage of the population being infected than with common or seasonal viral infections. As a result, the risk of a manufacturing process being exposed to the coronavirus was considered to be higher than usual, depending on the process. Therefore, most drug manufacturers moved quickly to perform risk assessments to evaluate the established good manufacturing practices (GMP) controls in their manufacturing facilities with respect to the new potential contaminant. The expectation of risk mitigation was also later reflected in a COVID-19-specific guidance published by the U.S. Food and Drug Administration (FDA) that stated "as a consequence of the COVID-19 pandemic, drug manufacturers were also expected to perform these risk assessments to ensure that product quality and safety, including employees' safety, are not impacted due to the pandemic" ***(1)***.

Fortunately for the industry, and for most of the manufacturing operations that would be considered at risk, careful implementation of GMP principles and practices are likely to protect product against the SARS-CoV-2 virus that causes COVID-19, as well as mitigate the risk caused by asymptomatic SARS-CoV-2-infected personnel. Some of these include:

- General hygiene measures and procedures in case of illness, feeling sick, open lesions, or sneezing and coughing
- Procedures for thorough handwashing and subsequent hand sanitization before entering production areas, and regular glove disinfection during production
- Use of environmentally controlled rooms within GMP-classified areas with specific gowning requirements (e.g., cleanroom gowns, overalls, lab coats, gloves, mask, gowning procedures) based on the operation performed
- Restriction of access to production areas to authorized personnel only
- Frequent cleaning cycles of the manufacturing areas and process equipment
- Design of general process equipment to ensure manufacturing areas are as closed as possible to prevent the potential entry of contamination
- Use of closed systems and, where product is exposed, use of isolators, restricted access barrier systems, and HEPA-filtered laminar airflow to protect the product
- Frequent cleaning cycles of the area and of the processing equipment to further prevent contamination from spreading
- Environmental and personnel monitoring to ensure strict adherence to GMP processes
- Use of air cascades and pressure differentials to allow air to flow from critical to less-critical areas
- Production flow in facilities designed to allow efficient movement and control of materials, personnel, products, equipment, and waste to prevent contamination
- Use of gamma-irradiated, single-use process equipment interconnected with aseptic connections and typically designated as functionally closed systems
- Stainless-steel equipment processes typically being cleaned and connected, followed by an in-situ steam sterilization cycle

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- Process venting points outfitted with hydrophobic-membrane filters that act as barriers to potential ingress into the process

Additional considerations are also taken when working with mammalian cell lines:

- Industry is expected to conduct adventitious-agent testing of a typical panel at the end of production cell bank. Additional risk assessment may need to be considered for banks manufactured after 2019.

- Measures to prevent and avoid viral cross-contamination during processing are well established, including deactivation and removal of viruses. These measures are likely to prove effective against COVID-19 as well. The large size of the COVID-19 virus makes it a prime candidate for removal by a 20-nm viral filter. Reports of the stability of COVID-19 over a wide range of pH values creates some potential concern around the effectiveness of a low-pH step for viral deactivation; thus, relying fully on this step for clearance is not recommended ***(2-4)***. Further studies in this area will enhance confidence in the effectiveness of the current processing steps. Alternatively, detergent or solvent deactivation is effective ***(4)***. Clearance of viruses through protein-A chromatography-capture and/or anion exchange are both functions of viral particle properties as well as protein and solution conditions. Using a panel of viruses that incorporates the physical characteristics of a coronavirus provides assurance of viral safety.

All these measures reduce the likelihood of an impact to product quality and safety due to pathogenic contamination to acceptable levels.

Learnings

The emergence of a pandemic or similar disaster must mobilize industry to assess the continued ability of their processes and quality systems to protect product quality and patient safety. Industry must be prepared to rapidly perform such assessments in the future when a similar situation emerges. In the case of SARS-CoV-2, manufacturing processes employing strict GMP adherence in their design, personnel controls, and the environments in which they are performed likely present an extremely low risk for possible SARS-CoV-2 product contamination. In most cases, when conforming to existing GMP, no additional measures are necessary to protect product and patient. In the event of an employee operating within the GMP manufacturing environment testing positive, a specific risk assessment should be performed to identify, analyze, and evaluate hazards that might need to be considered based on the specific circumstances.

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Q 1-2: How do you develop a mRNA/gene therapy drug product rapidly (Part 1)?

Background

Using messenger RNA (mRNA) to encode proteins for vaccination or protein replacement received its first in-vivo validation in 1990, when Wolff et al. demonstrated the production of a target protein after intramuscular injection in mice ***(1)***. The exact advantages of using mRNA products were discovered later, including:

- Allowing human post-translational modification of protein products with the potential for less immunogenicity and full functionality
- Permitting multimeric proteins that are not amendable to production in a bioreactor to be translated, folded, and assembled correctly in the patient's cells
- Producing transmembrane and intracellular proteins and trafficking them to their appropriate site in the cellular environment ***(2)***

Learnings

The preparation of plasmid DNA (pDNA) is the first step in the production of mRNA, where functional synthetic mRNA is obtained by the in-vitro transcription of a pDNA template. The pDNA template is transcribed into the mRNA mixture containing RNA polymerase and nucleoside triphosphates. Then, a cap may be performed during the reaction, or it can be added enzymatically post-transcription. Capping improves translation initiation by recruiting translation-initiation factors, thereby protecting the synthetic mRNA against degradation. Post-transcription, the pDNA template and bacterial DNA is digested by DNase. To be used as a drug substance, mRNA has to be purified to remove contaminating transcripts, which can be done by using a single chromatographic step that will separate the mRNA based on size and result in a pure, single mRNA product.

Two forms of mRNA are being studied for use in vaccines: nonreplicating (traditional) mRNA and self-amplifying mRNA. The traditional mRNA form presents two advantages: the molecule is quite simple and small, and the possibility of an unwanted immune response is lowered because there are no other proteins encoded. However, the traditional mRNA form is limited by its transient nature. The advantage of self-amplifying mRNA is that it performs self-replication and creates high levels of the antigen not found with traditional mRNA. The self-amplifying RNA vaccine was found to show a 64-fold lower dose required for an immunologic response compared to its mRNA counterpart ***(2)***.

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Q 1-3: How do you develop a mRNA/gene therapy drug product rapidly (Part 2)?

Background

Gene therapy products seek to insert genetic material into a patient's cells to treat or prevent a specific disease state. There are several types of gene-therapy products, which includes plasmid DNA, viral vectors, bacterial vectors, human gene-editing technology, and patient-derived cellular gene therapy ***(1)***. In much the same way, mRNA therapy products seek to insert mRNA material into a patient's cells to induce the production of a specific protein to treat or prevent a specific disease state.

The COVID-19 mRNA vaccine, for example, is made of mRNA and, to allow introduction into the cell, is encompassed by lipid nanoparticles. Once in the cells, it works by instructing the cells to make a protein that will trigger an immune response to COVID-19 ***(2)***. While there is no specific formula for the creation of a successful mRNA lipid nanoparticle (LNP) vaccine, there are items to keep in mind when designing such a product.

The mRNA LNP design is characterized by three main elements:

- mRNA sequence design and nucleotide modification choice
- Optimization of a LNP formulation to encapsulate and deliver mRNA
- Long-term storage ***(3)***

The first step in developing a mRNA therapy product is focusing on the API, as it is the most important part of the formulation. Considerations for designing or choosing the mRNA sequence should include the expression of encoded mRNA, immunogenicity due to recognition of mRNA as a foreign entity by intracellular RNA sensors, and mRNA stability. Nucleotide modification is equally important for this, as nonmodified mRNA molecules are recognized by the cellular RNA sensors, which nulls the effect of their expression. The translation efficiency and intracellular-mRNA stability can be increased substantially by mRNA capping. Additionally, the decay and translation efficiency are greatly affected by the untranslated region selection as well as the open-reading-frames sequence design.

The purpose of the LNP formulation is to design a nanoparticle that will shield the mRNA from degradation and enable efficient delivery into targeted cells. These formulations typically comprise a neutral structural lipid, cholesterol, a PEG-lipid, and an ionizable lipid. The ionizable lipid is incredibly important for enhancing the expression and provides the most-optimal immune response. The PEG-lipid component is helpful in increasing the shelf life of the solution by preventing aggregation. Considerations for PEG-lipids used in these formulations should take into account that, at high concentrations, they can prevent delivery of the mRNA.

The speed at which knowledge regarding mRNA vaccine development was gained during the pandemic was significant. This was a clear result of the unprecedented amount of global private and government funding that was made available for vaccine development.

Throughout the process of formulation, long-term storage and stability should be major considerations in design. The ultra-low storage requirement currently approved for mRNA vaccines due to instability at room temperature is a system constraint. To freeze LNPs, a cryoprotectant such as sucrose or trehalose should be added to the formulation. When added, these cryoprotectants preserve the physiochemical properties of the particles and allow maintenance of in-vivo delivery efficacy for three months.

Lessons Learned

In the case of a future pandemic, the availability of global and private funding to advance the development of research for a potential vaccine will again be critical. Governmental agencies globally should

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develop or revise plans related to the emergency funding of such research should the need arise. Where possible, governments should work together on such plans.

While the most desired mRNA vaccine would be one that could be provided in a lyophilized or a liquid formulation stored under refrigerated or room-temperature conditions, such formulations can take significant periods of time to develop, if even possible. Under pandemic conditions, when a vaccine is needed quickly, the initial use of an ultra-low storage condition should be considered as it removes many of the formulation challenges that would otherwise need to be overcome.

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Q 1-4: How did the route of administration of the vaccine impact the manufacturing operations needed and how did companies work together to allow rapid commercialization?

Background

The COVID-19 vaccine is considered a biologic parenteral drug formulated as a solution with the route of administration being sterile injections. Therefore, manufacturing of a parental drug is much more complex than the production of solid oral dosage forms such as tablets for oral ingestion. Apart from the significant differences in manufacturing a solution versus a tablet, there are many more controls required to ensure the product remains entirely sterile, including the control and use of sterile components such as vials and stoppers.

A vital step for these types of parenteral drugs that cannot be terminally sterilized is the aseptic fill–finish process in which solution is filled into containers, such as vials, and sealed with an aseptic closure, such as a stopper, as the final step before the product is packaged and shipped for patients. For several of the vaccine products developed, rapid freezing and long-term, ultra-low storage was also required. The formulation of a drug product can have a significant impact on the sterile fill–finish capabilities of the system. For instance, a product may require different sizes and composition of filters (membranes) to achieve sterilization of the product. Different characteristics of the drug product will also affect the parameters of the system used to fill the vials, such as flow rate and pressure. In many cases, the filling process requires specialized equipment due to specific product characteristics, such as solubility, viscosity, and the tendency of a product formulation to foam.

The fill–finish of millions of doses of vaccines is no small task; it requires significant manufacturing capacity, technical capabilities, and rigorous quality-control measures to ensure product quality. To achieve the volume of production needed, smaller pharmaceutical companies developing the vaccine teamed up with larger companies that had the capacity and technical expertise needed to produce it. Moderna, for example, utilized the sterile fill–finish capabilities of the contract manufacturer Thermo Fisher Scientific to aid in this incredibly important part of the manufacturing process ***(1)***. BioNTech partnered with Pfizer, an expert in the manufacture of sterile drug products ***(2)***.

Lessons Learned

One very valuable learning point during the pandemic was the importance and willingness of companies to partner with one another to rapidly develop the manufacturing process needed for vaccine production. In the event of a future pandemic, this will again be a crucial factor in the speed with which a novel vaccine can be commercialized.

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Q 1-5: Can vaccine manufacturing capacities be scaled up rapidly?

Background

Technology is rapidly evolving. Since 1990, when mRNA was first used in vivo, decades of research have gone into understanding mRNA–LNPs, so much was already known about mRNA before the start of the 2020 pandemic. Consequently, when the need for a COVID-19 vaccine arose, it did not take nearly as long as it would have if researchers were starting from scratch. Moderna's vaccine took only two days to finalize the sequence, 25 days to manufacture the first dose, and 63 days to be made available for the first patient to be dosed in a Phase 1 Clinical Study ***(1-3)***. Although the vaccine was made quickly, it was held to rigorous standards by the FDA as stated in *Guidance for Industry: Development and Licensure of Vaccines to Prevent COVID-19* ***(2)***.

To be effective, scaling up manufacturing of the novel drug product required a platform technology approach and available facilities in which to perform the operations. Such was the case with the COVID-19 vaccines.

Without considering process validation and regulatory approval timelines, the design, construction, and installation of a typical manufacturing facility requires a minimum of two to three years. In the case of COVID-19, however, the capacities needed to be scaled up much more rapidly. In 2010, the U.S. government started an initiative to fund sites that could be activated either for other vaccine purposes or to fulfill the needs of a pandemic-vaccine requirement. These sites were funded by the federal government to assure that they were built and able to operate without being a burden to the companies that owned the sites. Unfortunately, when these sites were activated, they were not able to fulfill the mission for which they had been created.

Based on the need to respond quickly, approaches were leveraged to bring new facilities online faster. The use of standard designs for cleanroom infrastructure by suppliers significantly abbreviated design timelines. These standard designs allowed the in-depth knowledge of all material components needed, the logistics to get the components to the site, and the site installation and qualification. The absence of the need for a design phase and the use of a repeat build and qualification strategy reduced normal facility construction timelines considerably. To accelerate the timeline further, standard designs of off-site, prefabricated, modular cleanroom infrastructures were used, as was the case in the Pfizer Kalamazoo site ***(4)***. These types of building modes not only halved the construction timeline but allowed them to be prequalified and used within a high-beam shell building, like a warehouse. This allowed the processing infrastructure to be added into existing shell buildings, with only slight modifications to building utilities, instead of constructing a new building.

Some companies, such as Thermo Fisher Scientific, actively worked to overcome this challenge and scaled up their manufacturing efforts quickly to meet the unprecedented demand. Initially, Thermo Fisher Scientific scaled up the manufacture of raw materials, relying on their current working supply of nucleotides and enzymes for early therapeutics development, which required no infrastructure changes. The company also built a 57,000-sq. ft. facility to manufacture the needed enzymes and nucleotides at industrial-scale levels using a platform approach to ensure a continued supply ***(4)***.

Lessons Learned

The pandemic situation emphasized the need for pharmaceutical companies and regulatory authorities to be prepared to encourage facility design, construction, and process installations that leverage and expedite innovation and the adoption of new technology with the goal of bringing the manufacturing facilities online rapidly.

For example, the use of modular facilities that allows manufacturers to continuously customize and reconfigure equipment to accommodate new products or processes could be encouraged. This use of portable modular manufacturing suites also provides new options for geographic diversification and facility reconfiguration.

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In addition, global governments should think about how to incentivize companies to design facilities in the future that are flexible enough to be repurposed during a pandemic if needed. This should be possible if the types of drug substance technology for the vaccine are not too different from the production techniques in use. The benefit of building flexibility into an operating facility is that the processes surrounding those manufacturing operations are upgraded over time including aspects of the quality systems. Building plants that are not routinely used and maintaining them in a hibernated state creates many unforeseen issues (e.g., equipment reliability, availability of trained staff, up-to-date quality systems).

A recent McKinsey report addressed the issue of being always prepared, with upfront investments over two years, followed by maintenance investments into the preparedness level. One of the issues addressed was the need for scaled-up vaccine facilities and having these facilities ready for use to supply 15 billion doses in six months. The investment over 10 years would be substantial, but the analysis also showed that the minimum economic loss globally would be 40 times higher if this investment was not made ***(4)***.

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II. People

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Q 2-1: How do we overcome the workforce shortage induced by COVID-19?

Background

The pandemic created an unexpected and prolonged shortage of workers. This shortage occurred because companies were neither prepared or equipped to ensure that workers could return to a safe environment. The initial worker shortage continued through the pandemic due, in part, to the impact of unvaccinated essential workers not being able to return to the workplace. Compounding this in some countries was government-augmented unemployment and housing benefits for workers, which created an economic driver for some vaccinated workers to delay their return to the workforce. The decreased availability of workers was largely also due to:

- Lack of available childcare causing parents to stay at home
- People voluntarily leaving the workforce entirely
- People going into early retirement
- People choosing to move from an essential job to a nonessential job

Low vaccination rates lengthened the severity of the pandemic as virus mutations (e.g., Beta, Delta, and Omicron) developed and moved through the population. This phenomenon of vaccine hesitancy was not limited to the United States but was also seen in many other countries where vaccines were available. In addition to the low vaccination rates, breakthrough infections of individuals who were vaccinated also resulted in a high infection rate and absence from work.

The pandemic has been emotionally challenging for many people and has changed many aspects of everyday life in unprecedented ways. One positive method to encourage employees and decrease worker shortages is to protect employees and create an atmosphere of safety. Companies began reimplementing certain aspects of their safety plans, such as wearing masks, social distancing, and frequent hand-sanitizing. Maintaining flexibility is important during transitional times while the right policies and support for workers are identified. Communicating with employees to ensure their needs are addressed is extremely important.

As a result of the worker shortages, companies began offering bonuses and other incentives to entice existing workers to return and to attract new workers during the pandemic. While this was effective, with the shift to remote working, it also created an opportunity for employees to change jobs for increased pay and presented opportunities outside their local areas without the need to relocate. Many called this phenomenon "the great resignation." Another form of enticement, once legally allowed, was a provision that permitted companies in the United States and select other countries the ability to mandate that workers be vaccinated and return to work or lose their employment. Some companies used a combination of requiring vaccination or weekly testing. This was an effective tool in that it incentivized many employees to take the vaccine and return to work.

Lessons Learned

To reduce the level of pandemic-induced worker shortages, companies should have plans in place to minimize employee risk in the face of potential pandemics, not only for an initial outbreak but also for the recurring waves of a pandemic that might be expected. The external environment should then be continually evaluated to determine what specific steps need to be taken to minimize the risk to employees, such as returning to specific requirements, like wearing masks and social distancing, that were implemented during the initial wave of the pandemic and then relaxed as the initial wave subsided. Ensuring that workers feel safe in the workplace is essential to maintaining the workforce.

As needed, there are also other approaches that companies can use to entice workers to return, thus reducing worker shortages. One such approach is the use of special bonuses to entice workers to return

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as well as to attract new employees to fill openings. To further encourage workers, companies should consider other added benefits such as more flexible work-from-home options (hybrid models seem to be the most accepted in the industry when nonessential work can be done from home a few days a week), childcare funding, company cell phones, and college tuition assistance. These were all used effectively by various companies as the pandemic progressed.

Companies may require disclosure of employee vaccination status, where allowed, to encourage workers to get vaccinated. In the United States, requiring the disclosure of COVID-19 vaccination status was allowed ***(1-2)***. Ultimately, companies may mandate that workers return to work or forfeit their employment if no remote work opportunities are available.

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Q 2-2: What restrictions should be considered for essential workers to ensure continued operation?

Background

At the start of the pandemic, few companies outlined what roles were considered essential for ensuring continued operations or what actions should be taken to protect those individuals.

Disruptions caused some companies to lose weeks and even months of manufacturing productivity as operations were shut down, limiting the spread of the virus. Even companies manufacturing the vaccine, such as Pfizer and Moderna, announced a worker shortage and difficulty recruiting people in Boston, Massachusetts ***(1)***.

The lack of personnel created a strain on the activities required to support production and batch release in the following areas:

- Lack of trained and qualified operators on the manufacturing lines
- Lack of trained technical support for setup and changeover
- Lack of qualified personnel for batch-records review

Without the required personnel, the operations will be interrupted and items like changeovers, technical support, and batch disposition cannot be provided or performed. Additionally, batch records cannot be reviewed, manufacturing processes are interrupted, product testing will not be completed, and product is held in the warehouse and cannot be released.

Lessons Learned

To prevent a strain on activities supporting production and batch release, sites must be ready to quickly apply new approved technologies (e.g., cameras to replace second-person verifications, implementation of electronic signatures for approval of batch records, augmented reality (AR) glasses or handheld devices for problem-solving on the line). New technologies, such as cameras and AR glasses, can support manufacturing-line operations by reducing the number of staff who need to be physically at the factory. Because of this, the qualified personnel can provide instructions remotely. The remote support could be used for such functions as line clearance, second-person approval, and on-the-job training. Dispensing of raw materials can be performed by having the second operator perform the verification remotely. A risk assessment would define what activities can be performed remotely and what activities cannot.

Automation of dispensing and filling lines can continue to be performed with a very limited number of resources on site and can ensure continued manufacturing using these principles.

Additionally, companies should have plans in place that outline and define what roles are considered essential for manufacturing operations. The plan should outline the emergency procedures to be implemented to protect those essential workers and account for any possible governmentally mandated regulations. While the exact characteristics of the infectious agent and its mode of transmission may not be known with all certainty, common-sense measures should be put in place, such as:

- Use of face coverings (if not already in place) in GMP areas
- Increased cleaning of facilities and equipment
- Nonessential workers working remotely
- Cafeteria, conference rooms, and other common areas closed, or the number of personnel allowed to be present restricted

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A review of how essential workers commute to work should be evaluated. Where the normal mode of transportation is via mass transportation (e.g., trains, buses, subways) or carpools, alternative modes of transportation to reduce the chance of workers being exposed should be considered. Plans to provide written documentation for employees to keep on their person for transit should be in place to comply with local, state, and federal requirements. All actions should be aligned with official government guidance (local, state, country).

During a company's evaluation of the risk, measures that may be more stringent than those outlined in official guidances may be warranted. Based on the critical nature of the products produced, a conservative approach should be used. Consideration should be given to scheduling groups of employees in shifts or units (i.e., having a defined group of people working together, while the other group is working from home and vice versa). If the shifts or units remain unchanged, the risk of the entire team being impacted is reduced. Personnel movement should be restricted to specific buildings or areas to minimize the possibility of a super-spreader event occurring. In some cases, personnel should be reassigned to minimize movement between buildings or areas.

Social distancing, visual cues of vaccination status, floor markers for the flow of personnel, and distance between individuals, should be implemented. If possible, electronic proximity detectors should be used to alert individuals when they are too close to one another. Physical barriers, such as plexiglass, should be installed to minimize exposure between individuals. Additionally, the speed of programs to screen workers on site should be established quickly (e.g., temperature, contact-tracing, quarantine requirements, daily health questionnaires). Considerations should be made during building design or renovations to ensure adequate air exchanges, and appropriate filtrations can be utilized if needed. In cases where the facility has not been designed with this in mind, other changes should be considered, such as upgrading the type of filters used or adding local filtration units.

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Q 2-3: What actions can be taken to minimize the risk of spreading contagions in the workplace?

Background

At the onset of the pandemic, the steps taken to minimize the spread of infection varied regionally. Companies referred to the guidance from public officials, world regulatory authorities, and regional health departments in an attempt to minimize the spread of COVID-19. In many cases, workplaces were temporarily closed, or the occupancy was limited to try and slow the spread of the infection in the population. Some companies also limited the operations to only essential workers, with supporting personnel working from home.

Some workers were classified as "essential" to the continuation of critical functions for society, and thus were required to be physically present to conduct business activities. Individuals with functions directly related to the manufacturing, testing, warehousing, and distribution of critical supplies, such as medicines, medical treatments, and medical supplies (including raw materials and test reagents), were classified in this way and continued to enter the workplace throughout the pandemic. Many workplaces instituted measures to try to minimize the spread of infectious agents. Workers who were deemed "nonessential" could, in many cases, continue to conduct their responsibilities virtually and were only permitted access on site when deemed absolutely necessary.

Lessons Learned

Manufacturers of pharmaceutical and medical-device products already have robust contamination controls in place for their manufacturing areas. These controls (e.g., gowning, high airflow and exchange, documented cleaning, disinfection programs) are already effective strategies to reduce the risk of spreading contaminants in the manufacturing area. A greater risk of transmission is in the office space and other communal areas, where controls are not commonly in place to minimize the spread of contaminants within the area or between personnel.

To be prepared for another such event, companies should have a health and safety plan for their office spaces and other areas that follows the guidance of the World Health Organization (WHO), U.S. Centers for Disease Prevention and Control (CDC), U.S. Occupational Safety and Health Administration (OSHA), or other regional government agencies. These plans should be based on risk assessments identifying contamination risks in the manufacturing process. A hazard analysis should be conducted as part of the development of the safety plan to identify where and how workers might be exposed to an infectious agent.

Work activities and common areas should be identified where employees come into close contact with others—for example, open-space workplaces, laboratories, meeting rooms, break rooms, cafeterias, locker rooms, and routes of entry and exit—where staff may congregate, or where a bottleneck might occur.

Actions that can be taken to help minimize the risk of spreading infectious agents in a workplace are outlined below. These actions should be taken as preventive steps at the beginning of the pandemic, even before the means of transmission is widely understood.

Engineering Controls

- Modify seat or workstation placement
- Construct physical separations, such as plexiglass partitions
- Install floor markers for flow of personnel with the required distancing between individuals
- Close or impose restrictions (number of allowable personnel present) on cafeterias, common areas, and conference rooms areas

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- Mandate the use of electronic proximity detectors to alert individuals when they are too close to one another (e.g., watches or other devices that will vibrate when workers get to within a specified distance of one another)
- Disinfect shared filing or storage spaces between uses
- Replace high-touch communal items, such as coffee pots or bulk snacks, with prepackaged single servings or encourage staff to bring in their own items
- Improve ventilation in the building and consider air-filtration systems
- Consider local filtration systems where the HVAC system cannot be improved or modified
- Ensure that adequate ventilation, air changes, and appropriate filtration can be used during the design of new facilities or those being renovated
- Open windows frequently in office spaces with windows that are not connected to controlled manufacturing facilities and keep doors open for as long as possible in office spaces without windows

Administrative Controls

- Mandate face coverings
- Require staff to stay home if they or a family member at home are ill; consider requiring testing before a return to site is permitted
- Limit maximum occupancy in common areas such as breakrooms
- Add additional shifts or stagger shifts, start times, and break times to reduce worker density
- Restrict personnel movement to specific buildings or areas and/or segregate working groups to avoid unnecessary interaction of personnel to minimize the possibility or impact of a super-spreader event
- Reassign personnel to minimize movement between buildings or areas
- Provide personnel training on correct masking, hand sanitization, and social distancing
- Implement programs to screen workers upon entering the site (e.g., temperature check, quick evaluation of health status using polymerase chain reaction (PCR) or antigen, and completion of daily health questionnaires)
- Establish infection notification, contact-tracing, and quarantine requirements
- Institute a cleaning and disinfection program for common areas and office areas
- Clean daily high-touch surfaces, such as door handles, desks, light switches, faucets, toilets, workstations, keyboards, telephones, handrails, printer/copier machines, and drinking fountains
- Post instructions and reminders at entrances and strategic places in the building on masking, hand sanitization, and cough-and-sneeze etiquette

Consideration should also be given to sanitation concerns, such as hand hygiene and surface disinfection. The strategic placement of hand-hygiene stations should be reviewed to reduce the spread of

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contaminants. Hand-hygiene stations should be placed at the entry and exits of high-traffic areas and in communal areas. Employees should be encouraged to sanitize their hands regularly.

The cleaning and disinfection of communal areas and workspace surfaces should be reviewed to ensure that it is adequate and effective against contagions of concern. The frequency of disinfection should be based on risk. For example, high-contact points (e.g., door handles, communal equipment, common areas) are likely to require more frequent disinfection compared to a dedicated office or workstation. Proof of cleaning and disinfection of common areas, workstations, and desks should be made evident with use of status cards or color indicators. The cleaning and disinfection regime should be made available to any personnel with interest or concern.

Having a cleaning and disinfectant program in place is not enough. The products used should be efficacious. Product efficacy against a specific virus can often be provided from the biocide supplier, health regulatory authority, or other governmental agency, for example, the Environmental Protection Agency in the United States. As the SARS-CoV-2 was a novel virus, none of the biocides on the market had label claims for efficacy against the virus at the beginning of the pandemic. However, the U.S. Environmental Protection Agency issued a list (List N) of all products that the agency expected to kill the SARS-CoV-2 coronavirus (COVID-19) when used in accordance with label directions until the disinfectant claims could be obtained ***(1)***. In cases where a new infectious agent is detected for which the biocide supplier does not have data, it is important for companies to look at other sources to identify if a specific biocide is expected to be effective.

When selecting products for use in an office setting, efficacy is of high importance and, during a pandemic, virucidal efficacy continues to be a key consideration. In addition to efficacy, consideration should be given to the format, health and safety requirements, material compatibility, and residues left behind by any disinfectants that are used. As these products are likely to be used on many different surfaces and material types, it is important to have a full understanding of how the biocides are to be applied (e.g., impregnated wipe, spray-and-wipes) and the material compatibility of the products to prevent adverse effects to surfaces (e.g., electronic touch pads, metals).

Requiring nonessential-worker vaccination was not commonly considered by companies until the COVID-19 vaccines were made widely available. Assuming that a vaccine is commercially available, before allowing workers to physically return to the workplace, it is recommended that the vaccine status of employees is cataloged for business continuity and safety purposes. Vaccine status is considered private information; however, in many countries, employers can require vaccine-status declaration without violating privacy laws by using either their company health care professionals or a privatized source. To improve their vaccine uptake rate, some companies provided space on site for health care professionals to administer vaccinations to employees.

Due to the long-term social-distancing already experienced by employees, it is recommended that companies discuss this with their employees and share the measures that will be taken to control the spread of infectious agents.

Employees and companies may prefer a staggered return to work. This can be achieved by surveying employees on when and how they would like to return to the workplace setting. Staggering the return of employees will also allow the company to evaluate the effectiveness of their control measures in place.

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Q 2-4: Who gets the vaccine first?

Background

As the first vaccines were developed and approved for emergency use, the number of doses available were limited. The question of who should receive the first doses was a topic of great discussion and debate. After health care workers and those at greatest risk, such as the elderly in long-term care facilities, came essential workers and those over 65. It was not clear during the vaccine roll-out exactly where pharmaceutical manufacturing personnel stood in regard to their priority in receiving the vaccine.

Lessons Learned

As the continued supply of pharmaceutical products is critical, it is important that the pharmaceutical industry workers within the manufacturing facility who are *directly* supporting the manufacturing floor operations and whose jobs cannot be performed remotely should be on the list to receive the vaccine early, and they should be the first vaccinated within the company based on their importance to the manufacturing operations. The line of workers that comprise manufacturing-floor operations should include the technicians directly performing testing of the products, the warehousing and shipping personnel responsible for moving and shipping the product, and other individuals who *must* be present for the product to be produced, released, and shipped. These workers are essential to ensure product supply. Companies should have plans in place to quickly identify these individuals in the event of another pandemic.

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Q 2-5: How do safety and human resource policies change after a vaccine is readily available?

Background

As the pandemic progressed and vaccines became widely available, a significant percentage of the population refused to receive the vaccination. There were multiple underlying reasons for this, among them, religious beliefs, general fear of vaccinations, and fears specific to the COVID-19 vaccine due to the lack of long-term clinical data. Compounding these challenges was a preexisting and active anti-vaccination movement that caused many individuals to avoid receiving the vaccine based on incorrect safety information. A strain on the availability of vaccinated employees resulted.

Because the COVID-19 virus is highly infectious, the percentage of the population required to obtain "herd immunity" to the virus would be expected to be very high. Adding to the percentage required to achieve this is the question of the vaccine's protection against infection over time as well as the potential of reinfection over time (See **Q 3-5**). Based on these factors, governments have taken the unusual step of allowing employers to request the vaccination status of workers, apply restrictions and, in some countries, require vaccinations as a term of continued employment.

For many, this was a paradigm shift and a significant change in expectations between employers and employees.

Lessons Learned

When on-site workers or visitors test positive for SARS-CoV-2, certain preventive measures need to take place. It is important to have processes in place for contact-tracing to let other employees know that they have been in contact with this individual, that they are at risk, and that they need to quarantine as well. The infected person should stay at home until they are no longer infectious. The duration may vary depending on the strain. Companies should follow the quarantine durations established by their public health authorities. The work area where this person was located needs to be cleaned and disinfected with an effective disinfectant. People identified through contact-tracing may need to provide a negative COVID-19 test result prior to returning to work. Personnel should be trained on masking, hand sanitization, and social distancing to lower the risk of future outbreaks and potential super-spreader events.

Companies should put processes in place where legally permitted to restrict the access to individuals (employees and visitors) who have not been vaccinated. For employees who have chosen not to receive the vaccination, companies should consider requiring a PCR or antigen test no older than 48 or 72 hours to be shown or, in some cases, should seek to replace them with employees who have been vaccinated. The vaccination status of all employees and visitors should be verified, and the records of the verification should be kept on file. Any exceptions should be rare and specific restrictions should be applied.

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III. Quality

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Q 3-1: How should companies assess their risk ahead of their next regulatory authority inspection?

Background

During the pandemic, the majority of global regulatory authorities eliminated most on-site inspections. Those that were conducted focused on new-product approvals, "high-risk" production sites, or inspections required to facilitate emergency authorization of COVID-19-related products. Because of this, the routine inspection schedule to which many organizations were accustomed did not occur.

During this period, two other factors impacted manufacturing facilities and their traditional operating model:

1. Manufacturing facilities that were involved with COVID-19 vaccine production were required, in some cases, to rapidly scale up facilities, people, and processes in order to respond to the pandemic. The speed at which this was required created circumstances that made it difficult at times to operate in full compliance with all current expectations regarding quality processes and procedures.

2. The change in the staffing model—to have all employees work remotely except those deemed essential for on-the-floor manufacturing operations, combined with an overall shortage of workers—created challenges to the manufacturing process quality systems that led to an increase in nonconformances to established processes and procedures.

Lessons Learned

Companies must remain aware of the actions that must be taken in their efforts to continue to supply the market with necessary pharmaceutical products. As operations return to a more normal, pre-pandemic state and in preparation for the return of regulatory authority inspections, companies should review the nonconformances and changes that have occurred and ensure there is adequate documentation to describe the event, the assessment made in its evaluation, the final decision, and any follow-up actions taken. Where there are gaps, identified action should be taken to eliminate or reduce those gaps. In many cases, this may include adding detail to the description, assessment, decision, and follow-up actions present in the original nonconformance. Any additions should be made as amendments to the original documentation. Where appropriate, if not already included, the documentation should clearly identify the pandemic as being the cause of or a contributing factor in the nonconformance. In addition, companies should perform a gap assessment of their current process against the applicable regulations and expectations. Any gaps identified should be closed or a plan should be established to close these gaps. These reviews and assessments, as well as the plans for closing any gaps, should be available for review during the regulatory authority inspections if needed.

Companies that are required to rapidly scale up, should perform a comparative assessment, if appropriate, against existing facilities, personnel, and processes to understand the similarities and differences in the approach used to bring the new capabilities online. Any differences should be evaluated and, where compliance gaps are identified, plans should be developed to close these gaps. These plans should be available for review during regulatory authority inspections if needed.

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Q 3-2: How will the "remote inspection" and/or "remote assessment" paradigm shift impact the industry?

Background

During the pandemic, regulators adapted their GMP and good distribution practices (GDP) inspection routines due to travel restrictions and safety concerns. In March 2020, the FDA paused most in-person inspections to safeguard its employees, according to a Government Accounting Office report. To continue to provide oversight, in many cases, the FDA began performing remote assessments ***(1)***. Similar reductions in on-site inspections were noted in the European Union with the relevant local agencies moving to more remote and hybrid inspections ***(2)***. Per regulations in place at the time of this publication, remote inspections of pharmaceutical manufacturing facilities by FDA were not permitted, as regulations require inspectors to be physically on site within the facility as part of any inspection.

This shift to remote assessments and inspections provided many advantages in terms of planning and flexibility but, as with all major shifts made within very narrow timelines, unanticipated challenges were encountered.

Lessons Learned

During the initial stages of the pandemic in 2020, it quickly became apparent that remote assessments and inspections required more thorough preparation than on-site inspections. From an operational perspective, some of the challenges GMP/GDP remote inspections and assessments can present are related to:

- Planning: Preparatory conference calls, connectivity tests, and a redundancy strategy are very important.

- Qualification Audits: Brand new manufacturing sites have no historical background on the GMP compliance level of the site being inspected.

- Communications: Cultural issues, poor clarity in requests from inspectors lacking local knowledge, and the quality of electronic connections can present obstacles.

- Online Tools: The use of different tools to share confidential documents, data, and information and use of mobile phones to capture live images from potentially hazardous manufacturing areas may be needed.

- Overall Conduct and Compliance to Agenda: Adapting to different time zones can be easily overcome; however, in many circumstances, the difference can lead to longer meetings.

As highlighted in the results of the European Federation of Pharmaceutical Industries and Associations' annual regulatory inspection survey, both industry and regulators are on a learning curve when applying virtual tools ***(3)***. From the experiences gained over the duration of the pandemic, companies will continue to benefit from investments they have made in facilities, equipment, and procedures for hosting live remote interactions with regulatory authorities. Nonetheless, firms may need to refine their hosting capabilities as they gain experience and as the approaches of global regulators evolve ***(2,4)***. In preparation for future pandemics, regulators should revise regulations and develop procedures that allow for a rapid shift to remote inspections if needed.

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Q 3-3: What cybersecurity changes have become necessary because of COVID-19?

Background

The increased level of digitalization in the pharmaceutical industry over the last decade has been unparalleled as computer-controlled processes, electronic batch records, information management systems, documentation and investigations systems, and the like have been implemented. With the increase in digitalization has come the increased risk of cybersecurity threats.

Due to the commercial benefits of new technologies developed during the COVID-19 pandemic, a surge of cybersecurity attacks directly related to acquiring COVID-19 information from regulatory bodies ensued. This affected both regulators, such as the European Medicines Agency (EMA), and private companies involved in the development of COVID-19 vaccines ***(1-2)***. Cybersecurity experts saw a 63 percent increase in cyberattacks related to the pandemic ***(3)***. From a human-error standpoint, even the change in environment has led to increases in successful phishing scams with a reported 47 percent of individuals being taken in by a phishing scam while working at home ***(4)***.

The rapid implementation of nonmanufacturing employees working remotely during the pandemic presented unanticipated challenges to corporate cybersecurity. A few of the many examples include:

- Instituting a "Bring Your Own Device," or maybe a better term would be "Use Your Own Device," approach enabled employees to access corporate data using their personal devices, which bypassed the standard "Corporate Owned Personally Enabled" approach.

- The use of new and, in most cases, rapidly evolving web-conferencing and information-sharing platforms has become commonplace. The emergence and rapid adoption of these new platforms was needed to keep pace with the proliferation of remote workers—a productive situation, but one that presented new cybersecurity risks.

- The vulnerability of home networks presented challenges as the security and reliability of these networks varied greatly.

- Sophisticated IT systems are susceptible to human error, especially systems that require a repetitive manual step like connecting to a company's virtual private network (VPN) before sending data. Simple errors can leave a system open to potential threats.

Lessons Learned

Based on the issues that arose during the COVID-19 pandemic, plans related to cybersecurity need to be developed so action can be taken early. Because situations can change rapidly, a strong cybersecurity plan should be in place long before a pandemic occurs and updated routinely. Likewise, risk assessments should be conducted that can direct priority actions, which should be reviewed regularly and updated with more pandemic-specific information as circumstances emerge.

Specific points relevant to a pandemic that should be considered for inclusion in risk assessments include:

- The expected increase in the number and complexity of cyberattacks related to external interest in new pandemic-related technologies the company may be developing

- Any changes in the way employees will work and interact with the company's electronic systems that may concern cybersecurity systems

- The current level of cybersecurity surveillance and its ability to rapidly identify cyberattacks based on expected changes due to the pandemic conditions

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- The preparedness of the company to respond to and contain a cyberattack based on expected changes in the way employees work and their locations (remote vs. in-person) due to the pandemic

- A review of the respective business continuity and crisis management plans should a successful cyberattack occur that shuts down critical systems, which should take into account the expected changes in the way employees work and their locations (remote vs. in-person) due to the pandemic

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Q 3-4: What new data-integrity risks are present because of a more remote workforce?

Background

Many work activities shifted from in-person to remote access because of the pandemic. This trend continues even though workers have been allowed to return to the workplace. This hybrid mode of working involves the application of various information and communication technologies (ICT) that are used for gathering, storing, retrieving, processing, analyzing, and transmitting information. It comprises the variety of software and hardware that drives ICT, for example, smartphones, handheld devices, laptop computers, desktop computers, video cameras, and wearable technology. ICT enables the remote work environment that allows employees to perform activities using remote access, and many of its components act as temporary or long-term depositories of data.

The increase in remote work activities utilizing ICT presents an inherent increase in risk as workers have moved from company-owned office equipment and a structured environment to home offices where the company has less control over the ICT. This increase in the use of ICT by remote workers adds more layers of data-handling and storage which, in turn, increases the risk of potential data-integrity issues. The risk to data integrity has been further impacted by the increasing number of audits and inspections taking place remotely; with these, a high number of external personnel and stakeholders require access to critical data, each using their own ICT, adding to the complexity. Companies have faced numerous challenges as a result of the sudden transition of having to assess—in many cases after the fact—potential risks and the available mitigation strategies available to address them.

Lessons Learned

Based on these challenges, some companies need to proactively evaluate the ICT currently in use within the company or those being planned to enable both in-person and remote work. A set of readily available ICT solutions approved for use by a company's IT units should be readily available and updated on a periodic basis. Senior leadership needs to ensure adequate resources (headcount and funding) are available for this activity. For this to be effective, it must be applied not only within the company but also to cross-company activities.

For example, prior to a remote audit, assessment, or inspection where external entities are involved, an agreement on mutually approved ICT solutions should be in place to minimize the risk of data-integrity issues. While this requires additional effort prior to the event, it greatly reduces the risk of issues occurring during or after the event.

Finally, in situations where paper-based systems remain (e.g., parts of or even the entire pharmaceutical quality system (PQS), additional attention should be placed on changing them over to electronic systems. Where that is not possible or practical, developing a process for how to ensure paper-based systems can be effectively performed remotely will minimize the potential for data-integrity issues, for example, how the ICT and related procedures generate the PDF-type documents to implement appropriate e-signature solutions.

Additional Reading

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Q 3-5: How can companies ensure optimal performance of business continuity plans during a pandemic?

Background

As an immediate response to the pandemic crisis, business continuity plans (BCPs) were activated, and their specific actions were implemented in order to shorten recovery time and mitigate the corresponding impact on respective pharmaceutical operations and activities. For many companies, the level of disruption created by the pandemic had not been envisioned in their existing BCPs. For example, few BCPs for companies with multiple manufacturing and warehousing locations envisioned a set of circumstances where all locations would be significantly impacted at the same time or one where the majority of personnel would need to work remotely. The situations and responses the BCPs detailed and how those related to the issues that occurred during the pandemic is a critical issue that needs to be assessed to ensure their future applicability in similar situations.

Prior to the pandemic, the development of detailed and executable plans in BCPs varied greatly among companies. In some cases, very robust BCPs were in place and, in other cases, their scope and detail were very limited; the need and requirements for BCPs were primarily driven by the business owner. For specific areas of BCPs, robust plans were required based on expectations from regulatory agencies and insurers. In the case of some regulatory authorities, a limited checkbox exercise was conducted in regard to the BCP and product supply. Before the pandemic, these types of reviews were logical; BCPs were not expected to be in place for a global disruption no one could have foreseen.

One interesting point related to the designs of prepandemic BCPs is that their structures differed greatly. Some were holistic and well-integrated across functions; many were not, with each function having its own standalone BCP. Some were focused more on the health and safety aspects, while others were aimed more on recovery of the IT infrastructure. These different BCP designs had a direct impact not only on the speed of response, but on the overall effectiveness of the response of the various pharmaceutical organizations.

Also, BCPs impacted whether the activities of the affected corporate organizations were disrupted or if they were able to implement the already formalized actions. It identified the need to conduct mock-up actions prior to the actual challenge in a defined frequency, providing the opportunity to the effectiveness of the arrangements in place, and to confirm that they remain robust and fit for use. This evaluation should be documented and justified.

As a result of the pandemic, the pressure put on the respective pharmaceutical organizations to develop or update BCPs had a direct impact on the overall activities performed. Even organizations that lacked BCPs or had very basic, though outdated, plans were forced to develop actions to ensure manufacturing continuity under pressure while developing or upgrading their BCPs. This led to testing BCPs in the field, which were developed, matured, and implemented in a very short period of time. This was by no means an efficient method, nor one developed with a full understanding of all the potential impacts it might have on the process, on other parts of the organization, or on larger operations.

Lessons Learned

While BCPs cannot cover all situations, they are critical in providing contingency plans for what actions to take when unexpected events occur. In the case of the pandemic, companies that had robust BCPs in place were better prepared to address the unexpected and unplanned situations that were not covered while implementing the actions for those that were. Companies that had not prepared had to build robust BCPs and test them in the field as the events were occurring, which is never the desired state as it may result in unanticipated consequences.

The pandemic provided a clear understanding of how critical BCPs are for an organization and individual companies. BCPs need to be holistic and, while different functions may have their own BCPs,

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they all need to fit together into one overarching BCP. For regulated aspects of the business, the BCPs need to consider how the requirements will be met for the actions being implemented, for example, how the requirements of the PQS will be met to ensure GMP compliance. BCPs must have built-in flexibility and adaptability, where appropriate, as well as a defined governance structure to oversee its implementation. The fact is that all remediation actions applied at every step, apart from ensuring the health and safety aspects, must sustain the respective GMP-compliance levels. This dual scope of compliance and adaptability needs to be highlighted in each step involved.

The recommendations below are designed to fill gaps about BCPs learned during the pandemic; the list is not intended to be all-encompassing, but these items should be considered. BCPs need to:

- Be developed, accessible, and ready for use
- Be detailed, reflective of current operations, and holistic
- Be flexible and adaptable with a defined governance structure to oversee its implementation; it is not possible to plan for every event, but it is important to have a plan for how to adjust the plan based on unforeseen challenges
- Take regulatory requirements into consideration to ensure the actions implemented continue to meet those requirements
- Take into consideration that the location of the workforce may be disrupted and the computer systems, hardware, and support that will be needed to enable it
- Include actions to be taken to keep employees safe in the event of a regional, national, or global outbreak of a pathogen (e.g., virus)
- Include detailed actions to be taken related to sustainability of suppliers, service providers, and the supply chain
- Be reviewed periodically and updated appropriately

Numerous references are available that can be of assistance when developing a BCP. One such document is standard*ISO 22301:2019: Security and resilience — Business continuity management systems — Requirements* ***(1)***.

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IV. Regulatory

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Q 4-1: How will regulatory authorities pivot between on-site and remote inspections?

Background

The COVID-19 pandemic necessitated a change in the world's working habits, and pharmaceutical industry regulators were no exception. To maintain regulatory oversight of both GMP and good clinical practice (GCP) activities, regulators worldwide adopted remote assessments, using video-conferencing software and devices to enable continuity of compliance oversight. Such technologies enabled remote inspections, evaluations, and assessments of certain activities and sites, even as the world was locked down with social distancing and travel restrictions.

To accommodate the changes made by regulators, manufacturing sites had to be prepared for desktop audits and/or remote interactive evaluations and remote audits, if requested. Tools (e.g., virtual glasses, translators) should be kept available should regulators wish to continue with these inspection methods in the future ***(1)***. Desktop audits (a high-level documentary review of policies and procedures) have been conducted prior to the COVID-19 outbreak; however, FDA has created new tools that can be used in combination with the desktop audits to better evaluate the sites and products in scope.

The FDA limiting unnecessary contact by conducting only prioritized domestic and foreign facility inspections and those that are deemed mission-critical ***(2)***.

When a planned inspection is not mission-critical, is not a prioritized domestic inspection, or is impacted by travel restrictions resulting from the public health emergency, the FDA will use other available methods and information to support regulatory decisions and the oversight of facilities, and it will supplement the inspection with other available tools. The FDA has developed a *Guidance for Industry: Remote Interactive Evaluations of Drug Manufacturing and Bioresearch Monitoring Facilities During COVID-19 Public Health Emergency* that describes the various remote interactive tools that the industry may request be used to conduct an evaluation. Within the guidance, any combination of these interactive tools can be used in a remote interactive evaluation. The FDA may ask to conduct a remote interactive evaluation prior to or following other types of regulatory oversight activities (e.g., an inspection or a request for records or other information) ***(3)***.

Lessons Learned

International regulatory authorities adapted their inspection approaches to ensure regulatory oversight of GxP activities, especially for high-priority products. Due to restrictions to protect public health, regulatory authorities used digital technologies, such as video-conferencing software and devices, to enable continuity of compliance oversight. The International Coalition of Medicines Regulatory Authorities (ICMRA) established the COVID-19 Working Group: Remote GCP and GMP Regulatory Oversight Inspections to review the adaptation of both GCP and GMP inspections during the pandemic to evaluate remote approaches. The team was chaired by the United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA) and had representatives of global regulatory agencies from the FDA, EMA, Health Canada, Swissmedic, HPRA Ireland, AEMPS Spain, ANSM France, PEI Germany, MHLW/PMDA Japan, TGA Australia, ANVISA Brazil, HSA Singapore, WHO, and Saudi-FDA.

The reflection paper from this working group presents the various factors that were considered by the inspectorates to assess the need for inspection on a case-by-case basis and apply established quality risk management principles and tools in their decisions on their regulatory oversight approach. This decision-making process has typically been performed by the lead inspector in consultation with operational management following discussions with the company being inspected. Factors considered by regulatory authorities during the pandemic have included:

- Local or international restrictions that have been in place to deal with the public health emergency
- The need to protect the health and safety of the inspector, inspectorate staff, and staff at the inspected facility, including reducing burden while dealing with the pandemic itself

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- The regulatory compliance history of the inspected; for example, considering if there have been repeat deficiencies identified or issues associated with information provided that could have potentially been misleading to an inspector
- The scope and objectives (preapproval, routine, or for cause) of the inspection
- The type of site (e.g., investigator, bioanalytical, sterile manufacturing) and intrinsic risks/suitability for remote assessment in order to fulfill the inspection objective
- The complexity of the activities of the site in terms of how challenging it may be to observe and adequately assess these remotely and by considering factors that may indicate regulatory compliance-related risks associated with the site
- How recently the site was inspected on site as well as the findings from that inspection
- Changes at the site since the previous inspection
- Activities delegated to contract research organizations (CROs) and service providers
- Accessibility of electronic systems owned and managed by service providers, CROs, and sponsors and whether these are accessible remotely owing to confidentiality rules
- Access to source documents (e.g., validation methods, process performance qualification data) while complying with applicable regulations and/or institutional policies
- Where redaction of documents must be performed to safeguard privacy and confidentiality
- The inherent risk associated with the activities conducted by the site, the types of products, and the need for the product (e.g., high unmet medical need)
- Ability of inspectors and the inspected facility to conduct inspections remotely considering resources, technical proficiency, and suitability of the site
- Whether the location of the site to be inspected made it more efficient or practical to use the remote approach to enable regulatory compliance verification ***(4)***

If the risk assessment determined that the use of the remote oversight approaches was not suitable for regulatory compliance activities, two approaches were typically taken. Either the inspection or regulatory compliance activity was postponed until the risk assessment changed or the pandemic conditions ensured that an on-site approach was suitable; or a hybrid approach was adopted. The primary factor that altered the initial risk assessment was the change in national or local rules to manage the public-health emergency, which typically resulted in the postponement of the inspection until the rules were changed.

The inspectorates have thus prioritized mission-critical domestic inspections, which might generally include preapproval, prelicense, surveillance, and for-cause inspections. The relevant agency will pre-announce such prioritized domestic inspections to the company to be inspected to help ensure the safety of the investigator and the firm's employees. This will provide the safest possible environment to accomplish the agency's regulatory activities, while also ensuring the appropriate staff are on site to assist the agency staff with inspection activities. In-person foreign inspections would be placed on hold as long as it was necessary and mandated by local pandemic restrictions ***(2–3)***.

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Based on the experiences of the authorities in 2020-2021, the ICMRA COVID-19 Working Group stated that remote inspections were an "enabling tool" that allowed the maintenance of "at least a minimal regulatory oversight during the pandemic, it is not the view of the group that remote inspections would fully replace an on-site inspection programme" ***(1)***. They explained that is because of limitations with the current technologies that hampered remote assessment of compliance. Using the remote approach to supplement typical in-person visits could be beneficial in post-pandemic times, but the extent of its use in situations outside of a public-health emergency would need to be evaluated.

Remote inspection tools have been adopted by several national regulatory authorities (e.g., MHRA, TGA Australia, EMA, Russia's Roszdravnadzor, ANVISA Brazil), though other regulatory agencies may not align with them. Companies should communicate in advance with the relevant national regulatory authority (NRA) about recertification inspections to learn if desktop audits and/or remote interactive evaluations will be acceptable. Until on-site inspections resume, manufacturing sites should discuss the determination of certification extensions with both local and national regulators if desktop audits and remote inspections are not feasible.

Manufacturers may also want to consider using the Mutual Reliance regulatory tool, where one authority can unilaterally rely on another authority's inspection reports, as promoted by the Pharmaceutical Inspection Co-operation Scheme (PIC/S). The WHO has been a major supporter of regulatory reliance and defines it as

> …the act whereby a NRA in one jurisdiction may take into account and give significant weight to assessments performed by another authority or trusted institution, or to any other authoritative information in reaching its own decision. The relying authority remains independent, responsible, and accountable regarding the decisions taken, even when it relies on the decisions and information of others ***(5)***.

The WHO *Guidance on Good Reliance Practices,* published in June 2021, is a comprehensive key document that elaborates on these principles further and provides greater detail on reliance in the "Quality" section of the document ***(5)***.

As part of trade agreements, Mutual Recognition Agreements (MRA) between the FDA and foreign NRAs allow drug inspectors to rely upon information from drug inspections conducted within each other's borders. Under the U.S. Food and Drug Administration Safety and Innovation Act, enacted in 2012, the FDA has the authority to enter into agreements to recognize drug inspections conducted by foreign regulatory authorities if the FDA determines those authorities are capable of conducting inspections that meet U.S. requirements ***(6)***. This concept has been used, and firms have received extensions of recognition based on these principles.

When an MRA or Mutual Reliance tool is employed, sites should propose to the NRA the use of the agreements to continue supply. Since the COVID-19 outbreak, MRAs have been used by several regulatory agencies that have resulted in waiving of audits.

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Q 4-2: How did global regulators collaborate during the pandemic?

Background

An unprecedented global pandemic of the twenty-first century required international regulatory agencies to take unprecedented collaborative actions. These collaborations played a momentous role in bringing the necessary countermeasures to the global population.

The ICMRA is an international executive-level coalition of key regulators from almost every region in the world. In 2021, it comprised 30 separate NRAs—including those of the United States, European Union, Canada, Australia, Japan, Singapore, and Brazil to name a few—with the WHO as an observer ***(1)***. The ICMRA provides a global strategic focus for medicine regulators and offers strategic leadership on shared regulatory issues and challenges. Priorities include a coordinated response to crisis situations.

During the pandemic, ICMRA played a key role and acted as a forum to support strategic coordination and to promote international cooperation among global medicine regulatory authorities. The aim of these activities was to expedite and streamline the development, authorization, and availability of COVID-19 treatments and vaccines worldwide. ICMRA members also worked towards increasing the efficiency and effectiveness of regulatory processes and decision-making ***(2)***.

During this period, other groups were also actively bringing together both industry and regulators such as the Regulatory Advisory Group (RAG) established by COVID-19 Vaccines Global Access (COVAX) and co-led by the WHO and the Coalition for Epidemic Preparedness Innovations (CEPI) ***(3)***.

Lessons Learned

The COVID-19 pandemic emphasized the importance of regulatory cooperation and information-sharing, which are essential elements of mutual recognition and mutual reliance. In early 2020, ICMRA emphasized the relevance of reliance in streamlining responses and facilitating efforts to multiple regulatory activities during public-health emergencies. For example, ICMRA and the WHO published reports, described agreements on common approaches from workshops, and discussed regulatory considerations related to the development of SARS-CoV-2 vaccine candidates and treatments among its members.

The following were some of the industry topics raised to regulators at the ICMRA workshop on *Enabling Manufacturing Capacity in the COVID-19 Pandemic* in 2021:

- Expanding reliance practices to include on-site inspection alternatives (e.g., virtual inspections)
- Adopting reliance practices for NRA-issued queries and associated manufacturer responses for supplements/variations
- Establishing collaborative review arrangements, similar to ACCESS and Project Orbis, for the review of supplements/variations for post-approval changes (PAC) of COVID-19 therapeutics and vaccines, as well as for non-COVID-19 products impacted by the pandemic
- Reducing the need for duplicative import and release testing ***(4)***

As many vaccines and therapeutics were rolled out beginning in early 2021, ICMRA further espoused the reliance principles among global regulators for efficient quality and manufacturing lifecycle management of these products ***(5)***. This included support for establishing a collective pharmaceutical quality knowledge management capability that could have the following benefits:

- Transitioning to harmonized, structured, and standardized electronic formats across regulatory authorities
- Sharing secure information about pharmaceutical manufacturing facilities, which can be contributed to, and accessed by, multiple participating regulators

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- Developing a framework that might, in time, support full harmonization of data elements submitted in the quality modules of the common technical document. This could pave the way for sponsors to make simultaneous submissions within a marketing authorization application to all associated NRAs and provide improved capabilities for both industry and regulators in the management of PACs

- Enabling more extensive mutual reliance among regulators by working to harmonize specific data expectations for sponsors and standards for review among regulators, so that regulators can be assured of the comparability of the assessments and related determinations of other regulatory authorities on whom they intend to rely ***(6)***

Beyond ICMRA, several regional collaborations among the regulators were also seen, such as in the Asia-Pacific Economic Cooperation (APEC) region ***(7)***. Several global organizations, such as the International Council for Harmonisation (ICH) and International Medical Device Regulators Forum also aided global collaboration and harmonization ***(8)***. There was also unprecedented regulator engagement with manufacturers developing pandemic vaccines and therapeutics from the beginning of the pandemic, as well as support offered in terms of regulatory expectations and abridged approvals.

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Q 4-3: How do you manage the reporting of deviations and other changes to the regulatory agencies during a pandemic?

Background

As a result of the first wave of the COVID-19 pandemic, there were many unexpected emergency situations for which many firms were not prepared. The ensuing staff and material shortages significantly impacted manufacturing operations.

After operations resumed, performing site-wide and organization-wide risk assessments with a cross-functional team helped many firms identify, group, and prioritize investigations. Deviations were resolved and reported based on their criticality and priority in terms of public health and manufacturing continuity. The existing knowledge on processes, products, quality by design, established conditions, quality metrics trends, and annual report information helped firms avoid reporting specific deviations and shortened the length of time needed for deviation investigations. The types of deviations companies encountered included a broad range of situations and areas impacted (e.g., suppliers, warehousing, manufacturing, quality, distribution).

Companies revised deviation procedures to adapt to the new normal of the pandemic in regard to criticality and priority in terms of public health and manufacturing continuity, reducing the burden on the internal quality function as well as on regulatory authorities. For the deviations or events identified as low risk, deviation reporting procedures and categories were revised to acknowledge (or log) and proceed or to document and proceed. These deviations were captured within the PQS. Some companies regularized specific events for short periods of time either through temporary change controls or by amending the procedures.

Deviations that were identified as moderate risk were in many cases reported as expected events and investigated with prior knowledge, requiring minimal new investigation activities. In general, if reporting was required, these deviations were reported in annual reports. Deviations that were identified as high risk were evaluated for their severity using existing knowledge where possible. If reporting was required, these deviations were generally reported as post-approval supplements.

To assist the industry, most regulatory agencies released guidances and Q&A documents or posted information on their websites related to deviation management and deviation-and-change reporting relief during the pandemic. These actions took time to implement, which created some challenges, especially during the initial phases of the pandemic.

Regulatory authorities eased up on PAC reporting about suppliers and manufacturing or control sites not listed in the existing regulatory filings, allowing for rapid implementation when it was critical to prevent or mitigate drug product shortages. For example, the EMA provided guidance under the Exceptional Change Management Process (ECMP) to cover regulatory strategies during pandemics to continue supply in the market. "The ECMP will permit the swift implementation of changes to suppliers and/or manufacturing/control sites necessary to reduce the risks of shortages under certain conditions intended to ensure the quality of the medicinal product while deferring the full assessment of the variation" ***(1)***.

Lessons Learned

Responding to a pandemic quickly and providing the flexibility needed in regard to deviation management and deviation and change reporting is critical for ensuring product supply. Several elements that should be considered are:

- Inclusion of the regulatory and quality functions in the BCPs

  The quality and regulatory functions should be included in the BCPs with a focus on the types of adjustments in processes related to deviations, as well as to deviation and change reporting, based on expected impacts caused by a pandemic. For many manufacturers, these areas were not a significant focus in past BCPs and, as such, plans were not readily available regarding the types of changes needed.

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- Rapid issuance or activation by regulatory authorities of temporary guidance and Q&A documents

  This includes specific information on changes in the regulatory reporting and handling of deviations.

  Most major regulatory authorities released temporary guidances and Q&A documents but, as was industry, they were not prepared for the speed at which changes occurred and industry's urgent need for direction from the regulatory authorities.

  Of particular importance is the reporting of changes in suppliers and manufacturing control sites not listed in the regulatory filings, as this can have a significant impact on product availability.

- Development of process to allow manufacturers to meet with authorities at an early stage when issues arise to understand regulatory expectations and propose innovative strategies for PAC submission and approvals

  While it is important for regulatory authorities to take broad actions, such as the issuance of temporary guidances, a process also needs to be put in place to allow the marketing authorization holder (MAH) to closely engage with those authorities early on about specific issues so they understand regulatory expectations and can propose innovative strategies for PAC submission and approval. This could involve such strategies as lower reporting categories, use of flexible submission strategies (e.g., ECMPs), reduced submission data requirements based on a risk assessment evaluation, and expedited assessments. Manufacturing operations within the pharmaceutical industry are diverse due to the diverse nature of the products; as such, situations will emerge during a pandemic that are specific and unique to any given product and will require unique solutions that can only be resolved on a case-by-case basis.

- Regulatory authority acceptance of temporary changes in quality controls

  MAHs facing difficulties performing the quality controls recorded in the regulatory filings, for example, due to a significant increase of manufacturing capacity to meet the demands of patients or other circumstances related to a pandemic, should contact the responsible regulatory authorities to seek temporary relief. They should be prepared to present an adapted control scheme, founded on a risk-based approach, with the goal of reaching agreement with the regulatory authority on the proposed strategy and the needed documentation to implement it.

- Regulatory authority acceptance of bundling changes for regulatory filings

  To reduce the resources needed to prepare, review, and approve individual submissions, regulatory authorities should allow for and encourage MAHs to bundle any changes, with the needed supporting rationale and data, to allow them to be implemented more rapidly.

- Regulatory authorities granting relief from specific routine reporting requirements

  MAHs, for justified reasons relating to a pandemic, may be unable to continue the expected full reporting requirements for specific items (e.g., full pharmacovigilance reporting, report of temporary disruptions/closures). In cases where the regulatory agency has not published relief through issuance of a temporary guidance or other document, the MAH should contact the regulatory authorities to explain the proposed temporary changes in reporting and request relief based on circumstances related to the pandemic. Relief may be in the actual reporting requirement or the method of reporting.

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- Regulatory authorities granting temporary acceptance status of specific anticipated deviations

  To assist the industry, regulatory agencies should consider granting acceptance status of specific types of anticipated deviations allowing companies to focus their resources on their manufacturing activities. A few examples are the drug supply chain security requirements waiver provided by FDA, extension of shelf life, or usage of expired product in clinical trials ***(2)***.

- Comprehensive risk assessment processes to determine deviation criticality and reporting

  Manufacturers should plan on carrying out a comprehensive risk assessment to evaluate the deviations being generated, allowing them to be divided into low-, medium-, and high-risk categories with decisions on reporting being made for medium- and high-risk deviations only.

  — Deviations classified as low risk could be handled as acknowledged (or log) and proceed, or document and proceed (e.g., missing process checks, second-individual in-person checking, annual reports filing, missing calibrations if equipment is found compliant in the subsequent schedule).

  — Deviations classified as medium-risk events may be grouped and addressed under a single deviation and, where required, reported to the respective regulatory authority as a grouped submission (e.g., FDA has provided clearance to, and faster approval time for, MAHs' filing for alternate lab and manufacturing sites for several applications grouped together).

  — Deviations classified as a major-risk category are handled case by case (e.g., change in source for an API, change in manufacturing site for vaccine or sterile-product manufacturing).

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Q 4-4: How can the industry self-ease the burden of handling deviations?

Background

As a result of the COVID-19 pandemic, some parts of the industry experienced an increase in deviations related to the quickly changing environment such as reduced on-site staff presence, supplier delays, and supply chain interruptions. The resulting volume of deviations, many of which had no impact on product quality, required companies to act and develop new processes for the evaluation, completion, and reporting of deviations with a focus on working to maintain product supply. In many cases, new processes were developed on how to approach and how to categorize deviations based on their impact to product quality.

Lessons Learned

Looking ahead, industry should have plans in place (BCPs) to quicky assess the anticipated impacts of a pandemic, especially regarding its impact on the volume and type of deviations expected, how deviations are handled, and how they are reported.

Some actions to consider that allow self-easement of deviation handling are:

- If a deviation is already open and the manufacturer can simply expand the scope of deviation to cover many similar changes in one deviation
- If minor deviations can be documented by the temporary change control to continue manufacturing operation
- If procedures that govern investigations covering discrepancies, deviations, and nonconformances are suitable during this public-health emergency or if they need to be updated
- If delayed and reduced testing can be implemented with sufficient data available (e.g., microbiological testing of environmental monitoring plates from low-criticality clean areas or routine stability testing supporting a marketed drug with a robust history of minimal degradation)
- If an on-site audit of a new or existing supplier is not possible due to pandemic-related travel restrictions, then using any measures that can be implemented (e.g., remote supplier audit, reassessment, or revision of the quality agreement so the qualification program can be updated)
- If the use of equipment changed or was reconfigured and it was not qualified prior to use to check the possibility of retrospective evaluation of equipment performance, if change control should be considered
- If servicing equipment requires a technician with special expertise and that individual is unable to be on site, are there alternative means to ensure that the equipment is suitable for use (e.g., when service that is normally performed on site was conducted with remote assistance)
- If a delay is acceptable in preventive maintenance or calibration activities for facilities or equipment impacting the function (e.g., adequacy for use) of facilities or equipment

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Q 4-5: How do you make sure that commitments are met, reported, and documented with deviations?

Background

Sites that are commercially approved for the release of drug products typically require a periodic re-certification to continue release of supply in some markets (as regulated by their NRA). During the pandemic, there have been instances where markets required an on-site inspection to recertify the site. Some NRAs conduct audits using a risk-based approach while others may schedule audits.

COVID-19 created profound disruption across the pharmaceutical industry in general, but it also impacted two specific areas of commitments from a regulatory and drug license prospective:

1. Regulatory Commitment—a specific description of how a company intends to manufacture and control the drug product that is defined in the dossier and is subject to regulatory action if changed after approval

2. Post-Approval Commitments—covers the elements of any agreement made with the regulatory authority during the drug review to fulfill after approval. For example, the FDA has a guidance for PACs that allows tracking of a new medicine's safety, efficacy, or optimal use. Similarly, commitments can be agreed upon from a chemistry, manufacturing, and controls (CMC) perspective as well as between the applicant and the regulatory authority during drug registration.

According to ICH *Quality Guideline Q12,* legally binding information considered necessary to assure product quality can be considered an established condition (EC). Therefore, any change to an EC requires a submission to the regulatory authority, similar to EMA variations ***(1)***. While the examples here are specific to EMA and MHRA, an applicant can reach out to the relevant NRA to request relief in registering changes if any deviation is observed for ECs based on the specific circumstances.

According to the *European Medicines Regulatory Network COVID-19 Business Continuity Plan,* variation can be delayed if it is in the interest of the applicant.

> Furthermore, it is exceptionally agreed for DCP/MRP/RUP, renewal and type II variation procedures to allow a "freezing" (holding the timetable at the same procedure Day and restarting it as soon as the response is received or as soon as the RMS AR is finalised) or "rolling back" (bring the procedure back to validation phase or back to clock stop with the usual timeframe of handling responses in the clock stop) of the procedure timetable due to unexpected and COVID-19-related capacity issues within the RMS, or when it is not possible for applicants to submit responses due to the COVID-19 pandemic ***(2)***.

All manufacturers are obligated to release batches with the Qualified Person (QP) declaration whenever major CMC changes are registered for a license, in keeping with EMA guidance EMA/196292/2014 ***(3)***. During COVID-19, certain exemptions were granted to submit QP declarations for post-approval commitments:

- Exceptional circumstances, when an on-site audit is not practical (e.g., atypical actives), are out of the scope of the declaration template

- An off-site, remote, or "paper-based" audit may be justifiable of benefit risk, but this can only be considered on a case-by-case basis

- In these cases, a suitable quality system is expected to be applied by the active substance and finished product manufacturers. As a principle, such controls must provide confidence that the active substance is fit for purpose and will not negatively affect the safety and efficacy of the medicinal product. The QP is expected to justify the controls in place on a scientific basis and record a risk assessment on a product-specific basis ***(3)***

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The MHRA allowed applicants the following flexibility to register post-approval commitments:

- Suspending the 30-day limit for replies to questions on Type 1B variations
- Waiving the requirement for leaflet mock-ups to be submitted to support variations; only text versions will be accepted and uploaded to the MHRA Products website
- Extending the permitted implementation period for label/leaflet changes following a variation from 6 months to 9 months ***(4)***

These scenarios may be limited to non-pandemic therapy, where applicants wish to delay registration of commitment due to resource constraints; however, there may be cases where pandemic-related therapies require accelerated approval. In such cases, using multiple tools to facilitate the implementation of manufacturing changes, such as risk-based reduction in supplement reporting categories and flexible assessment practices, may be allowed ***(5)***.

During the COVID-19 public-health emergency, the FDA has been willing to consider requests from applicants to submit certain changes using a lower reporting category based on such risk-mitigation information for marketing applications related to products in shortage or intended to diagnose, treat, mitigate, cure, or prevent pandemic ***(6)***.

Health agencies acknowledged challenges faced by applicants to register and manage PACs and deviation from commitments during the pandemic. Health agencies issued guidances and allowed some flexibility to manage and register these changes. Based on different scenarios, applicants were able to utilize a risk-based approach and then discuss applications with regulatory authorities to make sure that commitments were met and reported during the pandemic.

Lessons Learned

The importance of quick action by the regulators and industry to identify areas where flexibility is needed in regard to how regulatory commitments are met, reported, and documented cannot be stressed enough. While the actions taken significantly helped in working to minimize product disruption and minimize product shortages, the initial delay at the start of the pandemic created significant challenges. Developing processes to allow rapid implementation earlier, based on the knowledge gained during the COVID-19 pandemic, would be of significant benefit.

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Q 4-6: How do you prioritize submissions of changes in process during the pandemic?

Background

During the pandemic, the pharmaceutical industry was affected at many levels. MAHs for products against the SARS-CoV-2 virus (therapeutics, vaccines) had to prioritize submissions against unprecedented, shortened timelines, while those companies involved in non-COVID-19 products were met with a depleted workforce at the production sites, raw material shortages, and logistics delays due to cross-border movement restrictions. They also faced potential delays from regulators due to the shift in focus as a result of the pandemic and suffering similar restrictions in workforce mobility and availability.

Lessons Learned

For MAHs dealing with pandemic-related products, timing will be a highly critical factor in getting the different PACs through to submission and approved by the authorities in various jurisdictions. During the COVID-19 pandemic, in many cases vaccine MAHs adopted the grouping or bundling of product variations to reduce the number of submission packages. That, in turn, reduced the burden on regulators for individual assessments, aided timely approval, and ensured the maintenance of critical supply to the population. An excerpt presented at a regulators-industry workshop organized by ICMRA in 2021 shows the numbers of different PACs that were grouped in submissions for the different COVID-19 vaccines in the EU ***(1)***.

For MAHs of non-pandemic products, prioritizing changes based on the criticality of the change to ensure that an up-to-date and safe product can be placed on the shelves is recommended. Risk is best managed by coupling it with a scientific understanding of the process and product design for optimal product performance (e.g., as highlighted in ICH Q12, Chapter 2). Thus, risk-based assessments such as safety, integrity, strength, purity, quality (SISPQ) and safety, quality, integrity, purity, potency (SQuIPP) should be undertaken, and agreed-upon criteria should be documented in impact assessment tools that can be archived for inspections and Response-to-Questions submissions as needed by regulators.

For manufacturers with worldwide product supply and submission requirements, aligning with WHO guidelines on procedures and data requirements for changes and variations is recommended ***(2-4)***. Changes with no SISPQ/SQuIPP impact can be documented in the company's PQS ***(2)***. Minor changes, such as changes to the product composition, manufacturing process, quality controls, facilities, or equipment that have a minimal potential to impact the quality, safety, or efficacy of the biotherapeutic product, may be implemented by the MAH without prior authority review for biotherapeutic products ***(3)***. However, a notification to the authority within a specified time is recommended ***(2,4)***. The justification and supporting documentation for minor quality changes are not needed for such notifications, but the MAH should make them available to the regulatory agency upon request. Medium or major changes that need to be approved before implementation can be staggered and bundled in view of similarity or sequence of changes.

Additional novel tools that can be used for quick submissions and approval are:

- Matrix approaches to rationally reduce the number of lots required to support a manufacturing change for validation and stability (e.g., for vaccine combination products) can reduce the number of vaccine lots that are analyzed ***(2)***, which, in turn, can lead to faster submission and approval timelines and less destruction of lots that would reduce the vaccine supply

- Stability data approach and structured approach from ICH Q12, Chapters 8 and 9 ***(5)***

- Established conditions as part of ICH Q12 implementation ***(5)***

Combining these approaches for submissions that can be approved quickly according to pre-agreed change management protocols (or comparability protocols for the FDA) can enable multiple changes

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to be assessed and approved at the same time. For both pandemic and non-pandemic product scenarios, the MAH will need to provide a clear prioritization of changes, based on supply impact and predictive and strategic planning ***(6)***.

A best practice is for the MAH to clearly engage with the authorities at an early stage to understand regulatory expectations and flexibility based on the specific circumstances the MAH is facing.

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V. Supply Chain

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Q 5-1: Why have there been supply constraints for incoming materials, such as single-use systems, container closures, resins and filters, excipients, and APIs?

Background

During the COVID-19 pandemic, the demand for consumables and raw materials necessary for the production of vaccines and in need across the globe put a major stress on supply chains. Manufacturers of these single-use systems (SUS) and excipients were not prepared for fast turnaround times and large production; they were accustomed to consistent and repeatable orders and delivery. Lead times for consumables extended by up to 12 months, and lead times for sterile filters went from 2-3 months to 9-12 months ***(1)***.

The system as a whole was not ready for "emergency" supply chain management and, as demand increased, old methods became ineffective ***(1)***. Prior to the pandemic, the demand for incoming materials was increasing, especially plastic-based SUS and primary glass containers, such as vials and rubber components, a result of emerging applications in cell and gene therapies, monoclonal antibodies, and vaccines. Since the pandemic, local authorities have prioritized incoming materials for pandemic supply and established export bans on specific items. For example, the U.S. government has been using such measures as rated orders under the Defense Production Act to prioritize COVID-19 vaccine production, which primarily use SUS. That led to a very limited allocation of SUS to other biologics development programs and commercial manufacturing. The same applies to all drug substances, excipients, starting materials, intermediates, and devices.

Lessons Learned

Drug manufacturers and suppliers should work together on pandemic preparedness to assess demand and capacity, plan for supply manufacturing expansion, and act quickly on implementation. End users should understand their forecast and communicate with suppliers. Suppliers should anticipate the market demand, evaluate against their existing capacity, and fill in the gaps in a timely fashion.

Drug manufacturers should conduct a holistic risk assessment that includes alternative solutions, aside from safety stock; for example, they should anticipate validation of multiple sourcing on strategic components to increase their chances of meeting pandemic demand and maintaining continuity of supply for essential treatments.

There are a few reasons why lead times of materials were so severely affected, starting with restrictions in transportation globally and the movement of specific goods from vendors, which affected delivery times of critical raw materials or consumables ***(2)***. In the months leading up to the pandemic, the supply chain in the medical industry was growing far more complex around the world, with more third-party vendors competing for business with lower pricing and faster turnaround. This growing spiderweb of supply chains and the uncertainty of demand resulting from the pandemic accounted for delays that cascaded throughout the network, such as uncertainty of demand, inconsistency of supply, scarcity of material, delays in transportation, constraints in capacity, and shortage of labor.

The number of facilities that could meet manufacturing capabilities was limited, and the gaps in available workforce meant it took longer to hire and train qualified personnel to meet manufacturing needs ***(3)***. Several suggestions were made to mitigate the situation:

- Drug manufacturers should establish a cross-functional task force that includes procurement, supply chain, technical, quality, regulatory, and operations groups to respond to risks. The task force should be sponsored by corporate executives.

- Mitigation plans should include building safety stocks (wherever possible) and providing alternative supply solutions. Since all products and processes would be impacted, companies should prioritize based on a product's medical criticality and patients' usage volume.

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- Additional suppliers should be evaluated and qualified when deemed necessary. An emergency use guideline should be generated to allow faster turnaround of qualification.

Using the category of SUS as an example, some items to consider might include:

- Technical teams should categorize SUS using the product-family approach, evaluate component interchangeability, and use the available, functionally equivalent parts as a like-for-like change. Drug manufacturers and suppliers should systematically start working together to build a library of interchangeable components. This would facilitate the qualification process and shorten the overall lead time significantly.

- In some cases, drug manufacturers may be able to assemble and then steam-sterilize SUS in house. When the supplier has a limited capacity to perform SUS assembly, drug manufacturers should investigate if it could be done internally while maintaining the same level of quality.

- Currently, the shelf life of SUS post-gamma irradiation is typically 24 months, a conservative estimate that sometimes resulted in waste. Drug manufacturers and suppliers should work together to standardize the approach used to qualify shelf life with the aim of lengthening it. In the meantime, drug manufacturers and suppliers should collaborate to potentially extend the shelf life of recently expired SUS, based on scientific study or justification.

- SUS are designed to be used once and then disposed. When considering the reuse of SUS, the original technical and quality requirements should be met. Otherwise, SUS should not be reused.

Securing global pharmaceutical supply chains during a pandemic is a primary function of the pharmaceutical industry. Industry shares this responsibility with government agencies. Pharmaceutical companies must develop BCPs in order to maintain uninterrupted supply chain distribution during pandemics. BCPs provide a roadmap for senior management to understand and implement approaches to mitigate numerous potential risks related to supply in the event of a specific disruption.

Given the evolving nature of global pandemics, it is important to strengthen existing BCPs in order to ensure that they adequately address and assure robust supply chains that mitigate risk to ongoing product supply. Pharmaceutical manufacturers' BCPs were stressed during the pandemic, given the disruption to numerous routine operations and were often found lacking in detail and scope. Legacy BCPs developed strategies to react to multiple potential risks; however, these plans were typically related to individual risks and not scoped to deal with failures of multiple operational areas. Global supply chain activities were uniquely impacted during this time, and existing BCPs did not address the numerous risk factors encountered.

Pharmaceutical manufacturers remain responsible for developing strategies to mitigate supply-chain issues, which requires careful assessment of the interactions among the many existing business nodes. The interaction among logistics, manufacturing, packaging, warehousing, quality-control testing, product release, distribution, transportation routes, and importation activities are further complicated by segmentation of activities that are often owned by different entities at different stages of the supply chain. BCPs must be enhanced from the current state to include the learnings from the pandemic and work in synergy with BCPs of third-party suppliers to provide maximum assurance of an ongoing supply of pharmaceutical products. Several factors should be considered when deploying BCPs, among them, prioritization of medically critical products, product safety-stock levels, warehousing capacity, logistics, location of patient population centers, and third-party supplier robustness.

The BCP should detail stages of activity. For example, pharmaceutical manufacturers must act swiftly at the onset of a potential pandemic to ensure appropriate actions are taken immediately to limit the risk to product supply. This could include, to name a few, raising inventory levels to minimize the

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risk to manufacture and distribution, engaging third-party logistic providers to ensure BCPs are being activated, and inventorying available warehousing capacity.

The following items should be considered as part of a pharmaceutical manufacturer's BCP:

- Assess the need to increase inventory levels (commonly called safety stock) of materials used throughout the supply chain. Medically critical medicines should be prioritized; however, BCP risk mitigation should assess safety stock for all medicinal products.

- Assess the location where the safety stock is kept, as transportation might be impacted during a pandemic as well.

- Review safety-stock levels. Medical criticality and safety-stock levels should be assessed constantly throughout the pandemic to ensure necessary levels are maintained and changes in classification are fully understood and evaluated.

- Ensure alternative suppliers are qualified for all materials, as applicable, not only APIs, but even a lesser-priority excipient might become critical if it is missing.

- Assess medical criticality in each market as the supply needs may differ depending on local supply logistics, number of approved suppliers, and generic availability.

- Consider warehousing and distribution space needed for any increase in safety-stock inventory, including both cold-chain and controlled-room-temperature conditions.

- Consider the wider need for warehousing. This is especially important for cold-chain product where local intervention to provide storage for vaccine and other treatment may be given the highest priority.

- Consider locating finished product storage closer to the intended patient populations earlier in the product lifecycle. Moving the materials closer to the market prevents route and import disruptions from delaying treatment. Controls must be established to ensure robust approval and release processes to prevent unauthorized distribution prior to final product release.

- If third parties are involved, ensure controls are in place and are covered in existing agreements with third-party suppliers. The controls should be part of both internal and external company audit programs and should be detailed in procedures and signed agreements between parties. Where third parties are used, controls for product quarantine, release, and rejection procedures need to be fully understood. This is especially important if third-party warehouses will hold quarantined material pending release status change.

- Consider options for shipping lanes due to the real possibility of disruptions to shipping partners, potential changes in importation and customs procedures, port shutdown, protectionism, port-of-entry closures, and staffing shortages.

- Consider working with local governments to ensure transparent supply of materials and to make risk-based decisions that overcome disruptions, including deploying different ports of entry.

- Consider other supply options, including adding third-party suppliers and employing redundant sourcing options; pursuing these options during the initial stages of a pandemic may be appropriate. Given potentially lengthy approval times, consider building redundant options into routine operations. Redundant operations for the manufacture of medically critical medicines must be reviewed with consideration to building capability and gaining regulatory approval of local and regional sites of manufacture.

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- Assess stability beyond current shelf-life limits where possible. In the event of shortages during a pandemic, this data could be essential when working with local government authorities to allow a pandemic-related extension in the market where necessary.

- Identify markets that require import release-testing. Should there be a risk of shortages, firms should engage with local authorities to request waivers of import-testing requirements with justification based on supply dynamics. This could help offset local quality-control laboratory operations (including third-party laboratories), supply-chain impact, and customs and import delays that can be expected during pandemic periods, and the risk is very low for products where release-testing has already been performed.

- Consider the risk of counterfeits and product theft, which increase during pandemic times. BCPs should include a detailed plan for working with third-party logistic providers, warehouses, and distribution providers to ensure safe and effective transport of products with rapid notification and response over and above normal procedures.

- Consider building multiple BCPs to allow for variation; for example, firms may have one BCP for routine product supply and a separate BCP for clinical-trial supply.

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Q 5-2: What impact did cold-chain custody and manufacturing bottlenecks have on vaccine supply?

Background

Current supply-chain capabilities were assumed to be sufficient to effectively distribute the COVID-19 vaccines during the pandemic. This was not the case, however, as the current supply chain for vaccines is quite complex and was undergoing issues before the onset of the pandemic. As a result of globalization, supply chains have become increasingly complex. According to Dr. Robert Handfield from Poole College of Management and Dr. Daniel J. Finkenstadt from Naval Postgraduate School, there are several key challenges associated with vaccine supply chain: national security issues, personnel shortages, lack of coordination, shortage of supplies, limited capacity, vaccine damage, gaps for rural areas, and misinformation about vaccines ***(1)***. Specifically, cold-chain products are very sensitive to movement and high-touch handling. Processes had to be validated to prove that the vaccine was still effective after being handled in dry-ice packages. A narrow window existed between getting the vaccine out of cold storage and into patients before it deteriorated; this was a big challenge and a crucial step for vaccination of the population. Cold-chain issues presented an even greater challenge in rural areas than in highly populated cities, where the routes were longer and access points for citizens were more difficult. Consequently, vaccine inequality is more prevalent in lower-income countries that have a less-developed cold supply chain ***(2)***. Even in the United States, the capacity for temperature-monitored cold rooms could not handle the mass volume of vaccines that needed cold storage.

Lessons Learned

When redesigning or looking to improve the cold supply chain for events such as the COVID-19 pandemic, there are several key points to consider:

- Design distribution around freezer requirements. Consider planning the distribution of vaccines en masse as soon as they arrive in the district. Reduce storage time after arrival by hosting events for mass immunization. Be creative with distribution techniques, such as the case in Ghana where drones were used for deliveries ***(2)***.

- Optimize shipment size to limit requirements for cold-chain capacity. Transition packaging to smaller units by dividing bulk shipments into smaller packages shipped more frequently.

- Increase cold-chain storage capacity. Increasing freezer or cold-room capacity can help to ensure the cold chain can handle planned or unplanned incremental volume changes associated with vaccines. Increasing storage capabilities is important for mass immunization events and to protect against delays in vaccine distribution. Possible solutions for increasing cold chain include renting temporary warehouse capacity from the private sector, using portable freezers or mobile cold storage, or evaluating the cost effectiveness of repairing existing cold-chain units.

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Q 5-3: How have vaccine manufacturing supply chains become more vertically integrated?

Background

To get vaccine manufacturing up to the demand required during the COVID-19 pandemic, many companies, such as Pfizer and Merck, had to take on some of the responsibility of obtaining the consumables and raw materials that were required to manufacture their drug products. With extended delays from suppliers overseas, the need grew for companies to supply materials in house where they could control their own lead times.

Lessons Learned

Pfizer needed extremely large quantities of a cationic lipid that was essential to their COVID-19 vaccine, which was not readily available at the start of the pandemic. Consequently, Pfizer developed a process to create the lipid needed based on one that had already been devised by a Pfizer research and development (R&D) facility. Assembling a team of 50 people, they performed a tech transfer from the R&D facility to a larger manufacturing facility to create the cationic lipid needed for the Pfizer vaccine. To expedite the timeline, the team found ways to consolidate steps wherever possible and, soon after, the facility was producing 20 kg of cationic lipids each week. Instead of relying on vendors, Pfizer was able to control the output production of lipid that served their purpose in creating the COVID-19 vaccine ***(1)***.

Similarly, acquisition of companies that produced important raw materials needed for the creation of vaccines helped reduce lead times by streamlining communication between the two entities. Merck started production of a high-purity synthetic cholesterol product; the launch of this product followed Merck's acquisition of AmpTec, which combined AmpTec's mRNA technology with Merck's extensive expertise in lipid manufacturing.

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VI. Relevant Vendor and Supplier Resources

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